

Week 5: VPC Creation and Ec2 Instance Connection

◆ Step 1: Create the VPC

1. Go to **AWS Management Console** → **VPC**
2. Click **Create VPC**
3. Choose **VPC only**
4. Enter:
 - **Name:** Custom-VPC
 - **IPv4 CIDR block:** e.g. 10.0.0.0/16
5. Click **Create VPC**

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

Custom-VPC

IPv4 CIDR block Info
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block Info
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

VPC dashboard

✓ You successfully created vpc-06c619efe692a9b1c / Custom-VPC

vpc-06c619efe692a9b1c / Custom-VPC Actions

Details Info

VPC ID vpc-06c619efe692a9b1c	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-06bf6c63c20826066	Main route table rtb-084322c6e62720a4d
Main network ACL acl-02abe1c965ee632ae	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -

◆ Step 2: Create Subnets (Public & Private)

Create subnets in same **Availability Zones** for high availability.

Example:

- **Public Subnet:** 10.0.1.0/24 (AZ-A)
- **Private Subnet:** (AZ-A)

Steps:

1. Go to **Subnets** → **Create subnet**
2. Select your VPC
3. Choose AZ
4. Enter CIDR
5. Create subnet

The screenshot shows the 'Create subnet' page in the AWS console. The page is titled 'Subnet 1 of 1'. It has a 'Subnet name' field with the value 'public_subnet'. The 'Availability Zone' is set to 'United States (N. Virginia) / us-east-1a'. The 'IPv4 VPC CIDR block' is '10.0.0.0/16'. The 'IPv4 subnet CIDR block' is '10.0.1.0/24' with a note '256 IPs'. There is a 'Tags - optional' section with a key 'Name' and a value 'public_subnet'.

6.

The screenshot shows the 'Create subnet' page in the AWS console. The page is titled 'Subnet 1 of 1'. It has a 'Subnet name' field with the value 'private_subnet'. The 'Availability Zone' is set to 'United States (N. Virginia) / us-east-1a'. The 'IPv4 VPC CIDR block' is '10.0.0.0/16'. The 'IPv4 subnet CIDR block' is '10.0.2.0/24' with a note '256 IPs'. There is a 'Tags - optional' section with a key 'Name' and a value 'public_subnet'.

7.

◆ Step 3: Create an Internet Gateway (IGW)

1. Go to **Internet Gateways**
2. Click **Create internet gateway**
3. Name it `Custom-IGW`

Create internet gateway info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Custom-IGW

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	Custom-IGW	Remove

Add new tag

You can add 49 more tags.

Cancel Create internet gateway

- 4.
5. **Attach it to your VPC**

Attach to VPC (igw-01123e7c80df42b6e) info

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

vpc-06c619efe692a9b1d

AWS Command Line Interface command

Cancel Attach internet gateway

- 6.

◆ Step 4: Create Route Tables

You need **separate route tables** for public and private subnets.

Public Route Table

1. Go to **Route Tables** → **Create route table**
2. Select VPC

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
public-rt

VPC
The VPC to use for this route table.
vpc-06c619efe692a9b1c (Custom-VPC)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q public-rt X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

- 3.
4. Add route:
 - o Destination: 0.0.0.0/0
 - o Target: **Internet Gateway**
5. Associate with **public subnet**

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Add route

Cancel Preview Save changes

6.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
public_subnet	subnet-00b36ee6e12452101	10.0.1.0/24	-	Main (rtb-084322c6e62720a4d)
private_subnet	subnet-04d7f9a67adf0880d	10.0.2.0/24	-	rtb-07a0614d9c16386aa / private-rt

Selected subnets

subnet-00b36ee6e12452101 / public_subnet X

Cancel Save associations

7.

Private Route Table

- Keep default local route only (no IGW)

- Associate with **private subnet**

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
private-rt

VPC
The VPC to use for this route table.
vpc-06c619efe692a9b1c (Custom-VPC)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Name

Value - optional
private-rt

Add new tag
You can add 49 more tags.

Cancel **Create route table**

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
public_subnet	subnet-00b36ee6e12452101	10.0.1.0/24	-	Main (rtb-084322c6e62720a4d)
private_subnet	subnet-04d7f9a67adf0880d	10.0.2.0/24	-	Main (rtb-084322c6e62720a4d)

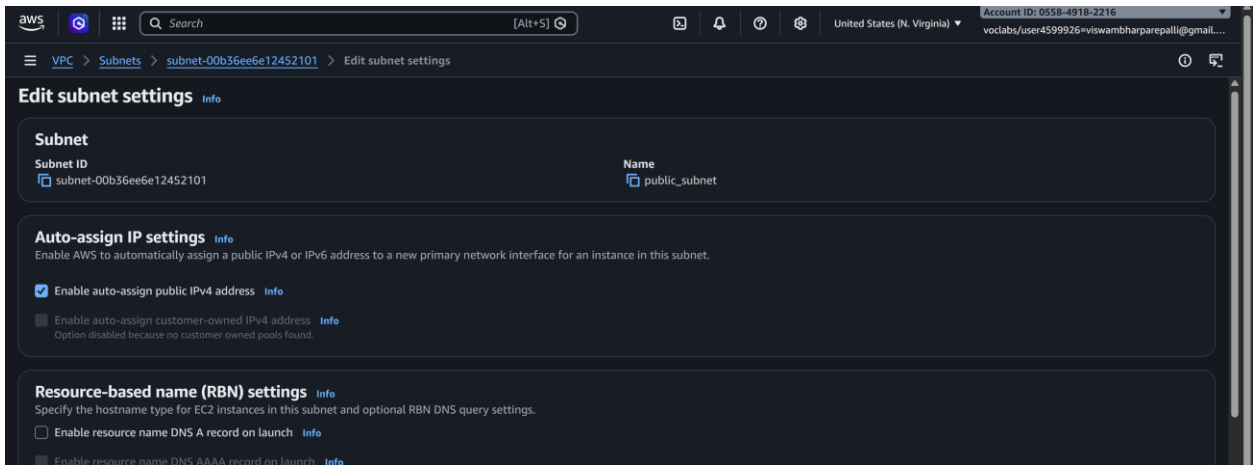
Selected subnets

subnet-04d7f9a67adf0880d / private_subnet

Cancel **Save associations**

◆ Step 5: Enable Auto-Assign Public IP (Public Subnet)

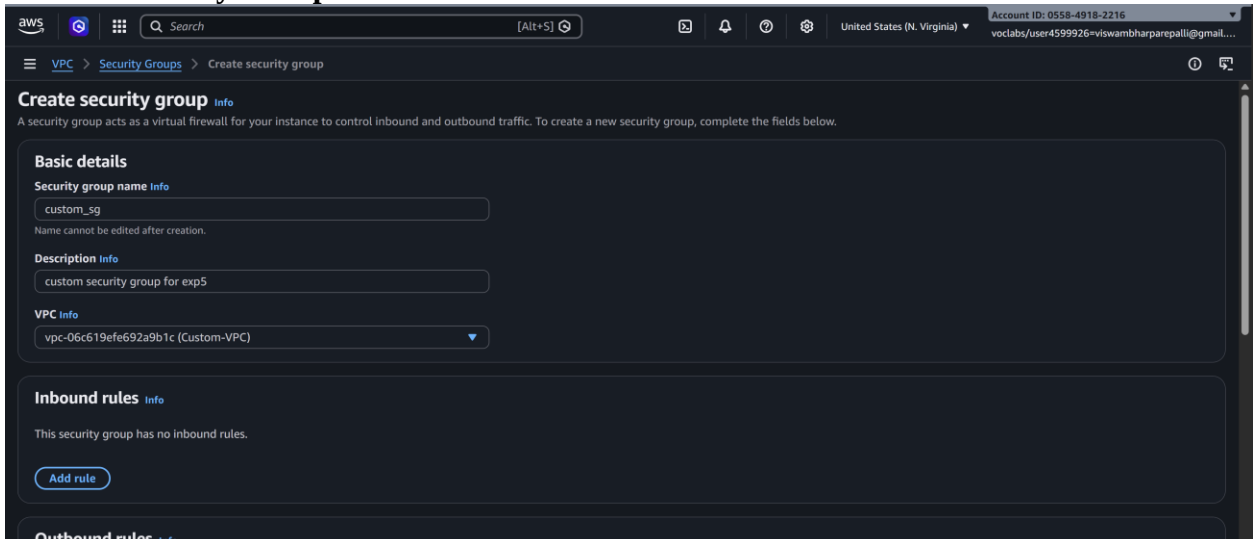
1. Select **Public Subnet**
2. Go to **Edit subnet settings**
3. Enable **Auto-assign public IPv4 address**

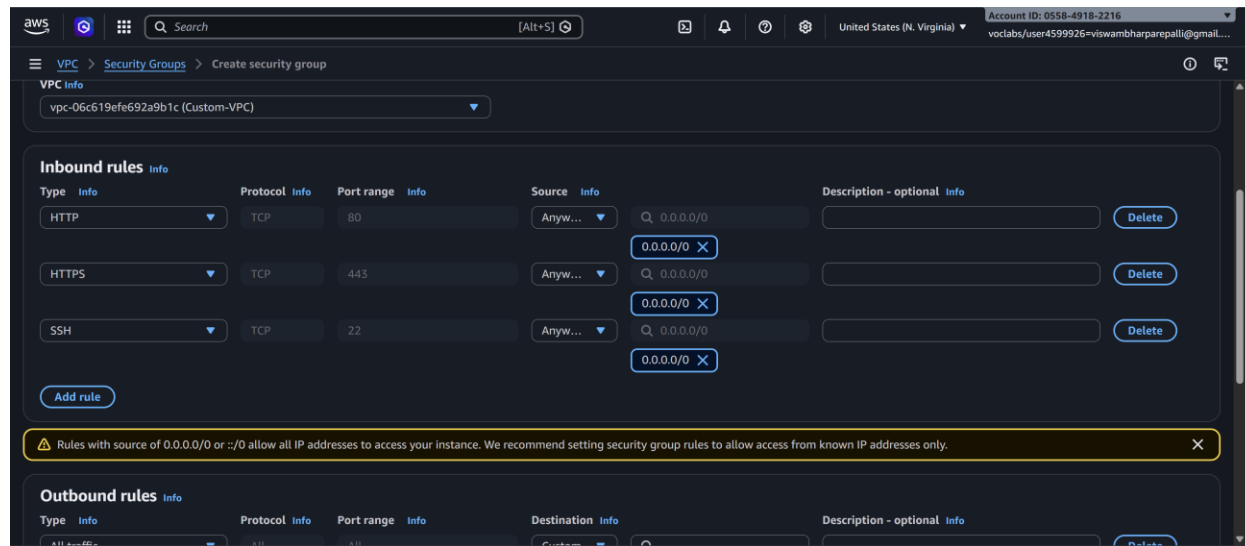


4.

◆ Step 6: Configure Security Groups

1. Create a Security Group

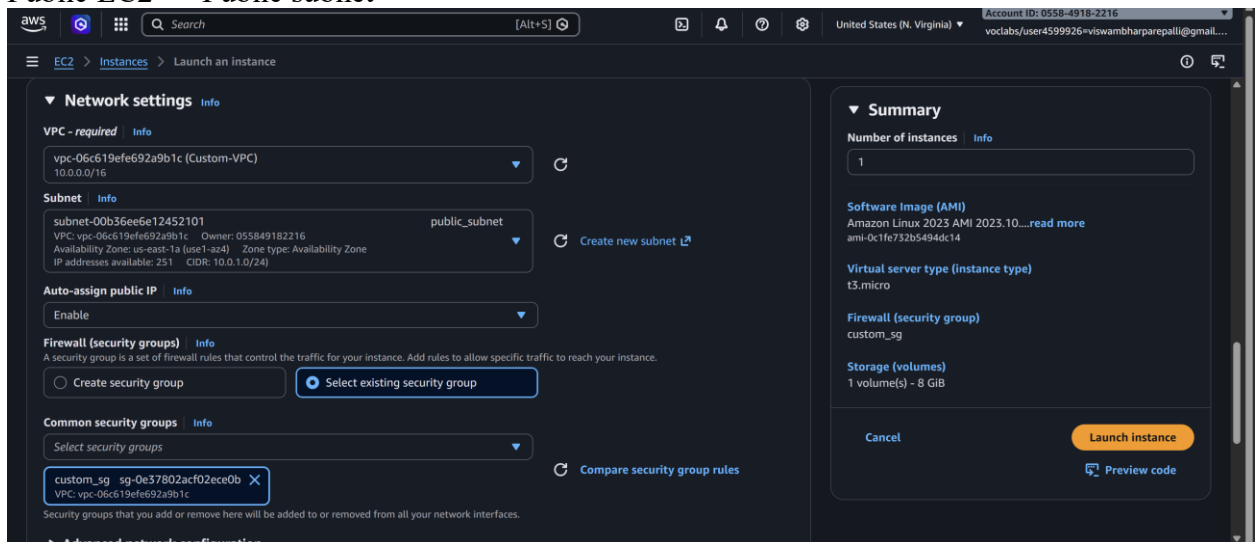




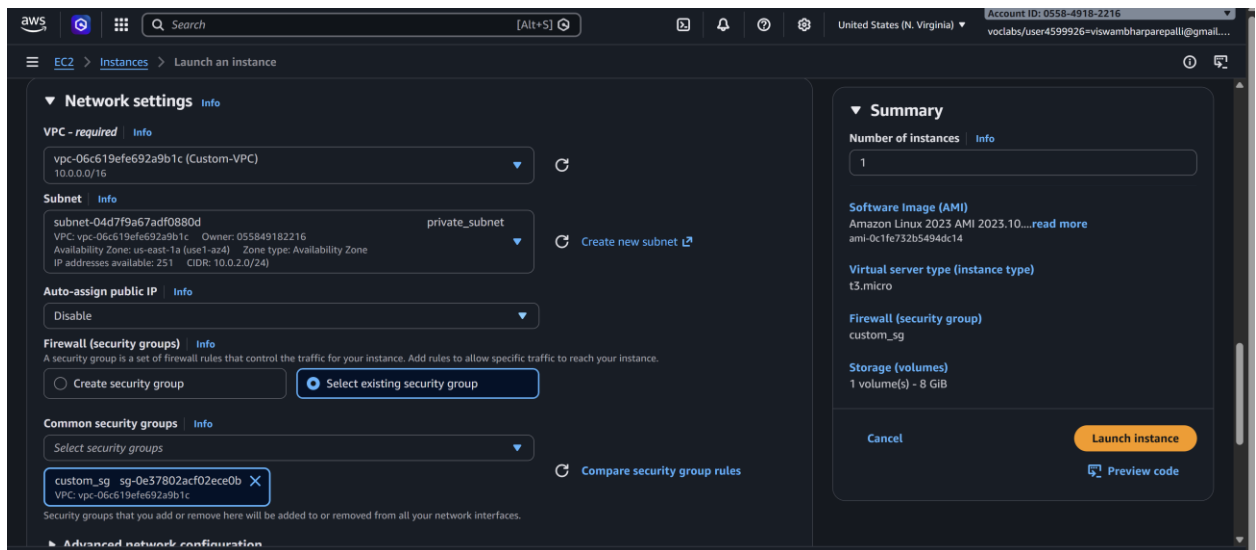
4. Attach to EC2 instances

◆ Step 7: Launch EC2 Instances

- Public EC2 → Public subnet

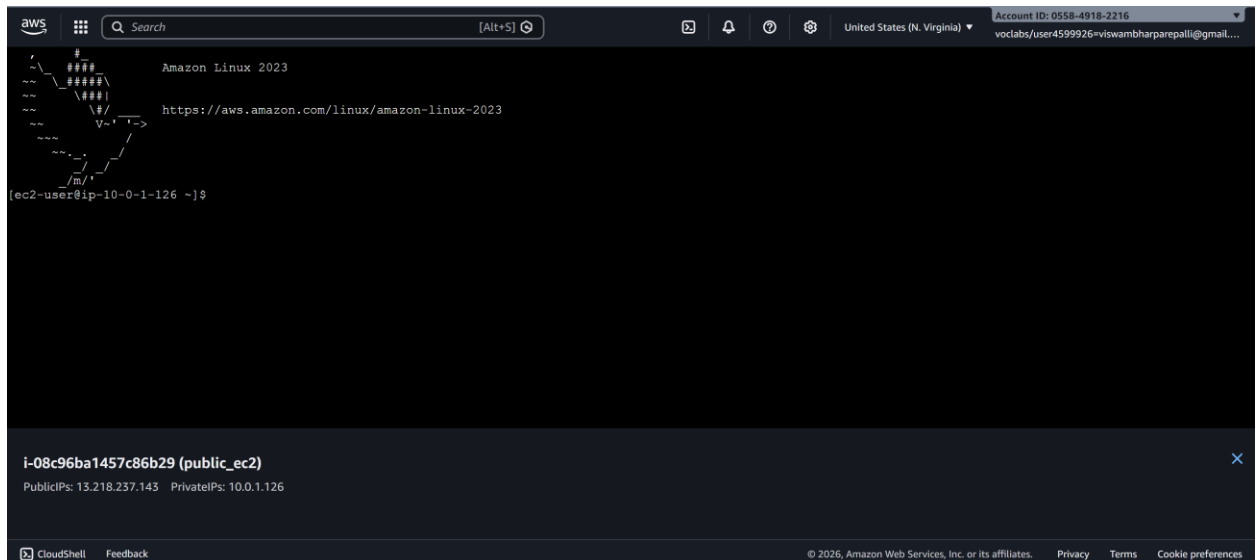


- Private EC2 → Private subnet
- Attach correct security groups.

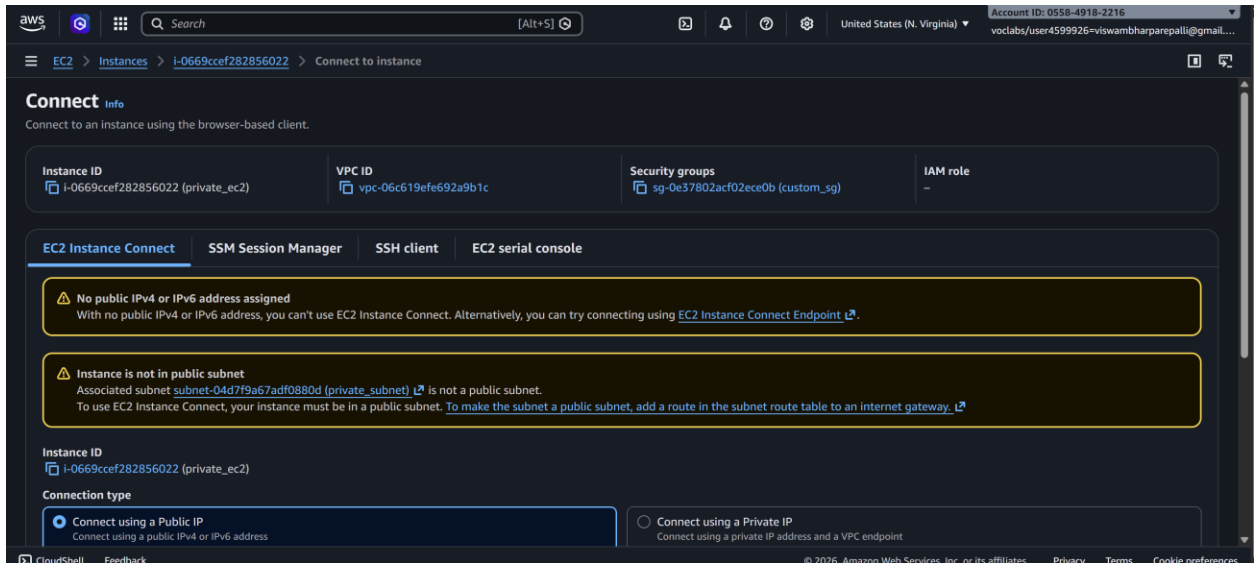


◆ Step 8: Test the Setup

- Public EC2 → Internet access ✓



- Private EC2 → Internet via NAT ✓
- Private EC2 → No direct inbound access ✓



Scenario Based Questions:

1. You have a web application where users access a website, but the database should not be exposed to the internet. How would you design the VPC? Which resources go into public and private subnets?

Answer:

Use a layered subnet design:

- Public subnets: Internet-facing Application Load Balancer, NAT Gateway, Bastion host (if needed).
- Private subnets: Web/App servers and RDS database.
Attach an Internet Gateway (IGW) to the VPC. Public subnets route 0.0.0.0/0 → IGW. Private subnets use NAT Gateway for outbound internet. Database stays isolated with no public IP.

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2. An EC2 instance in a public subnet cannot access the internet. What VPC components would you check and why?

Answer:

Check in this order:

- Internet Gateway attached to VPC.

- Route table: 0.0.0.0/0 → IGW.
 - Instance has public IP or Elastic IP.
 - Security group outbound rules.
 - NACL outbound rules.
 - Source/Destination check (if NAT/bastion).
- Most failures are missing public IP or wrong route table.
-

3. Your application server needs to connect to an RDS database securely. How would you configure security groups and subnets?

Answer:

- Place RDS in private subnets (no public access).
 - App servers in private or public subnets depending on architecture.
 - Create separate security groups:
 - App SG: allow outbound DB port.
 - RDS SG: allow inbound DB port only from App SG (not CIDR).
This enforces least privilege and avoids exposing DB publicly.
-

4. You are asked to design a VPC for 500 servers today, but it should scale to 2,000 servers in the future. How would you choose the CIDR block?

Answer:

Plan for future growth:

- 2,000 instances require far more IPs due to scaling buffers, ALBs, RDS, Lambda ENIs, etc.
 - Choose something like /16 CIDR (e.g., 10.0.0.0/16) instead of /24 or /20.
AWS reserves 5 IPs per subnet, so larger CIDR prevents painful re-architecture later.
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5. A company wants secure connectivity between its on-premises data center and AWS VPC. Which AWS services would you choose and why (VPN vs Direct Connect)?

Answer:

- Site-to-Site VPN: Fast to deploy, encrypted over internet, lower cost. Good for quick setup or moderate traffic.

- Direct Connect: Dedicated private link, predictable latency, high bandwidth, no internet exposure. Best for production hybrid workloads.
Often companies use Direct Connect + VPN backup.
-

6. You need to connect two VPCs, but both use the same CIDR range. How would you solve this problem?

Answer:

VPC peering and Transit Gateway don't support overlapping CIDRs. Options:

- Re-IP one VPC (best long-term solution).
 - Use NAT Gateway or proxy-based routing.
 - Use PrivateLink if only specific services need exposure.
Overlapping CIDRs are a design mistake — avoid them early.
-

7. Multiple microservices running in different subnets need to communicate securely. How would you design routing and security groups?

Answer:

- Keep default local route in route tables for intra-VPC communication.
 - Use security-group-to-security-group rules, not wide CIDR rules.
 - Separate SG per service (e.g., ServiceA SG allows inbound only from ServiceB SG).
 - Avoid opening entire subnet ranges unless absolutely necessary.
-

8. Design a VPC for a 3-tier application (Web, App, DB) with high security and scalability. Explain subnets, route tables, gateways, and security groups.

Answer:

Architecture:

- Public subnets: ALB, NAT Gateway.
- Private App subnets: Application servers.
- Private DB subnets: RDS.

Routing:

- Public route table → IGW.
- App subnet route → NAT Gateway.
- DB subnet → no internet route.

Security Groups:

- ALB SG allows 80/443 from internet.
- App SG allows traffic only from ALB SG.
- DB SG allows DB port only from App SG.

Use multi-AZ subnets for high availability.

9. You suspect unusual traffic inside your VPC. How would you monitor and analyze network traffic?

Answer:

- Enable VPC Flow Logs → CloudWatch or S3.
 - Use CloudWatch metrics and alarms.
 - AWS GuardDuty for threat detection.
 - Traffic Mirroring for deep packet inspection if needed.
- Flow logs are usually the first step.
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10. Why can't a private subnet have an Internet Gateway directly attached?

Answer:

Internet Gateways attach to the VPC, not individual subnets.

A subnet becomes “public” only when its route table sends traffic to the IGW. Private subnets intentionally avoid that route for security isolation.

11. What happens if route tables are misconfigured in a VPC?

Answer:

Traffic gets dropped or blackholed:

- Instances lose internet access.
 - Services cannot reach each other.
 - Load balancers fail health checks.
- Routing issues often look like application failures but are network-layer problems.
-

12. What happens if a route table has no local route?

Answer:

In AWS you cannot remove the local route — it is automatically added.

If hypothetically missing, resources inside the same VPC couldn't communicate across subnets.

13. An EC2 instance allows traffic on port 80 in the security group, but traffic is still blocked. What could be the reason?

Answer:

Common causes:

- NACL blocking inbound/outbound.
 - Wrong route table.
 - Instance has no public IP.
 - Web server not running or listening.
 - OS firewall (iptables/Windows firewall).
- Security group alone doesn't guarantee connectivity.
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14. You need to create a VPC that will host 1,000+ EC2 instances across multiple AZs. How do you decide the CIDR block?

Answer:

Estimate future IP usage, not just EC2 count:

- Include scaling buffers, load balancers, RDS, endpoints.
 - Use /16 or /17 CIDR to avoid fragmentation.
 - Pre-plan subnet ranges per AZ (e.g., /20 per tier per AZ).
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15. Only a company's corporate IP should be able to SSH into EC2 instances. How would you implement this securely in AWS VPC?

Answer:

- In EC2 security group inbound rule:
 - Allow TCP 22 from corporate public IP CIDR (e.g., 203.x.x.x/32).
 - Prefer bastion host or AWS Systems Manager Session Manager to avoid exposing SSH directly.
-

16. An EC2 instance can send traffic out but cannot receive responses. Which VPC component might be misconfigured and why?

Answer:

Likely issues:

- Network ACL blocking inbound ephemeral ports (stateless).
- Asymmetric routing.
- Security group missing inbound rule for return traffic (less common because SGs are stateful).
- NAT Gateway/instance misconfiguration.
NACLs are the usual culprit because they require explicit inbound and outbound rules.